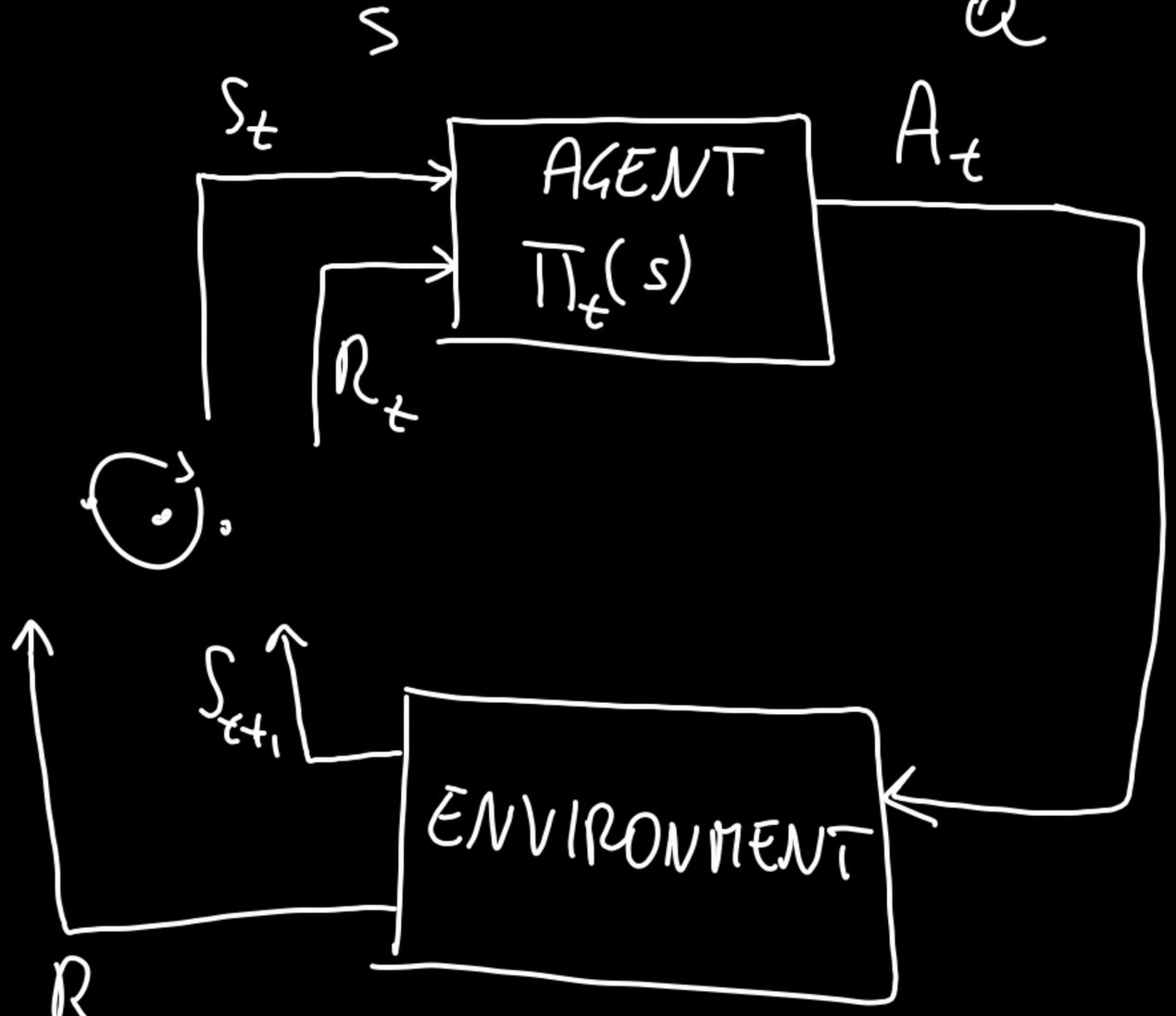
2/11/26 - FORMAL STUDY OF RL



$$S = \{s_1, s_2, s_3\} \quad R = \{r_1, r_2\}$$

SEQUENTIAL DECISION MAKING

$t = 0, 1, 2, \ldots$

$S_0, A_0, R_1, S_1, A_1, R_2, S_2$

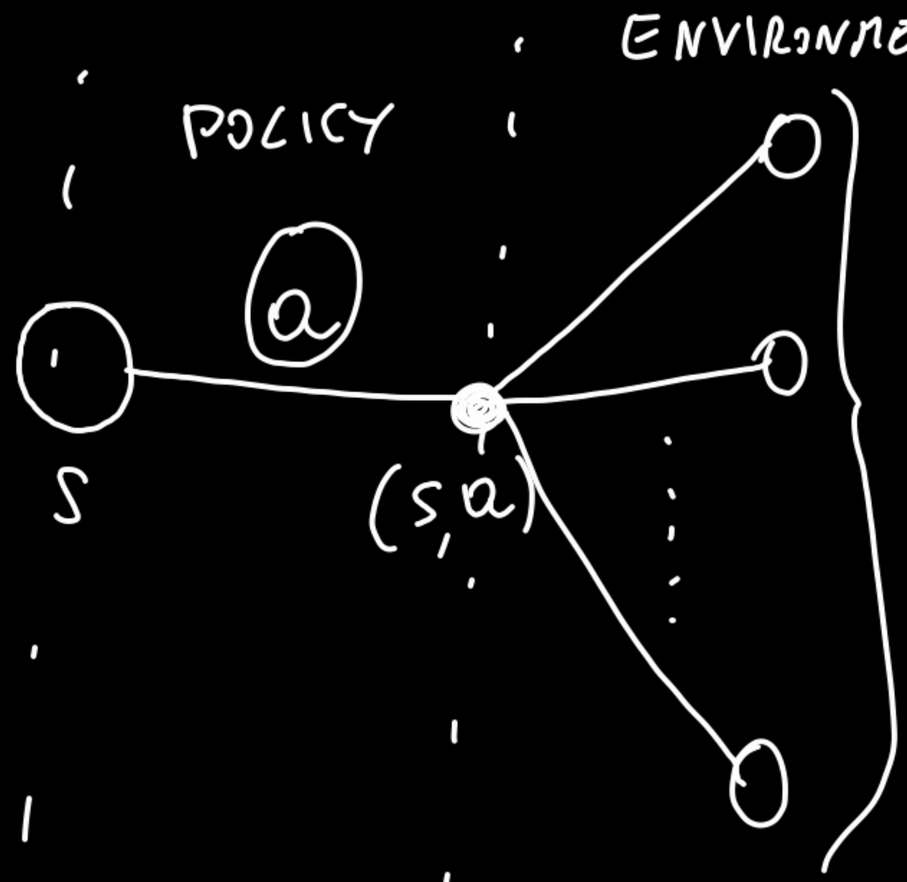
TRAJECTORY

FINITE MARKOV DECISION PROCESS:

S, A, R ARE DISCRETE AND WITH A FINITE NUMBER OF ELEMENTS.

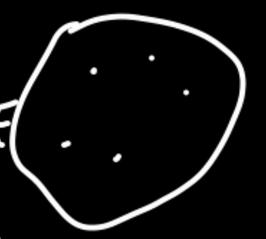
MODEL OF THE ENVIRONMENT.

PROBABILISTIC MODEL OF ENVIRONMENT.





- S IS THE SET OF ALL POSSIBLE STATES.
- A IS THE SET OF ALL POSSIBLE ACTIONS.



- $A(s)$ SET OF POSSIBLE ACTIONS FROM STATE S.
- R SET OF POSSIBLE REWARDS.

AGENT

ENV.

(s_1, r_1)

(s_1, r_2)

$(s_1, r_{|R|})$

s_1

MODEL OF ENVIRONMENT

$p(s', r \mid s, a)$

a

s

(s, a)

(s_2, r_1)

$(s_2, r_{|R|})$

s_2

$p(s_1 \mid s, a)$

$(s_{|S|}, r_1)$

$(s_{|S|}, r_{|R|})$

$s_{|S|}$

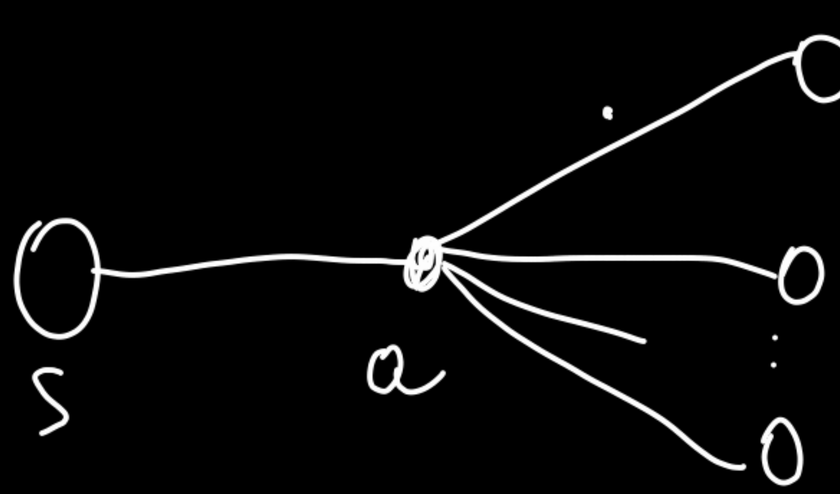
$\mathcal{R}$

MODEL OF ENVIRONMENT

$$\boxed{p(s', r \mid s, a)} \triangleq \Pr\{ S_t = s', R_t = r \mid S_{t-1} = s, A_{t-1} = a \}$$

IN GENERAL:

- UPPER CASE LETTERS → RANDOM VARIABLES
- LOWER CASE → SPECIFIC VALUES OF THE VARIABLES.



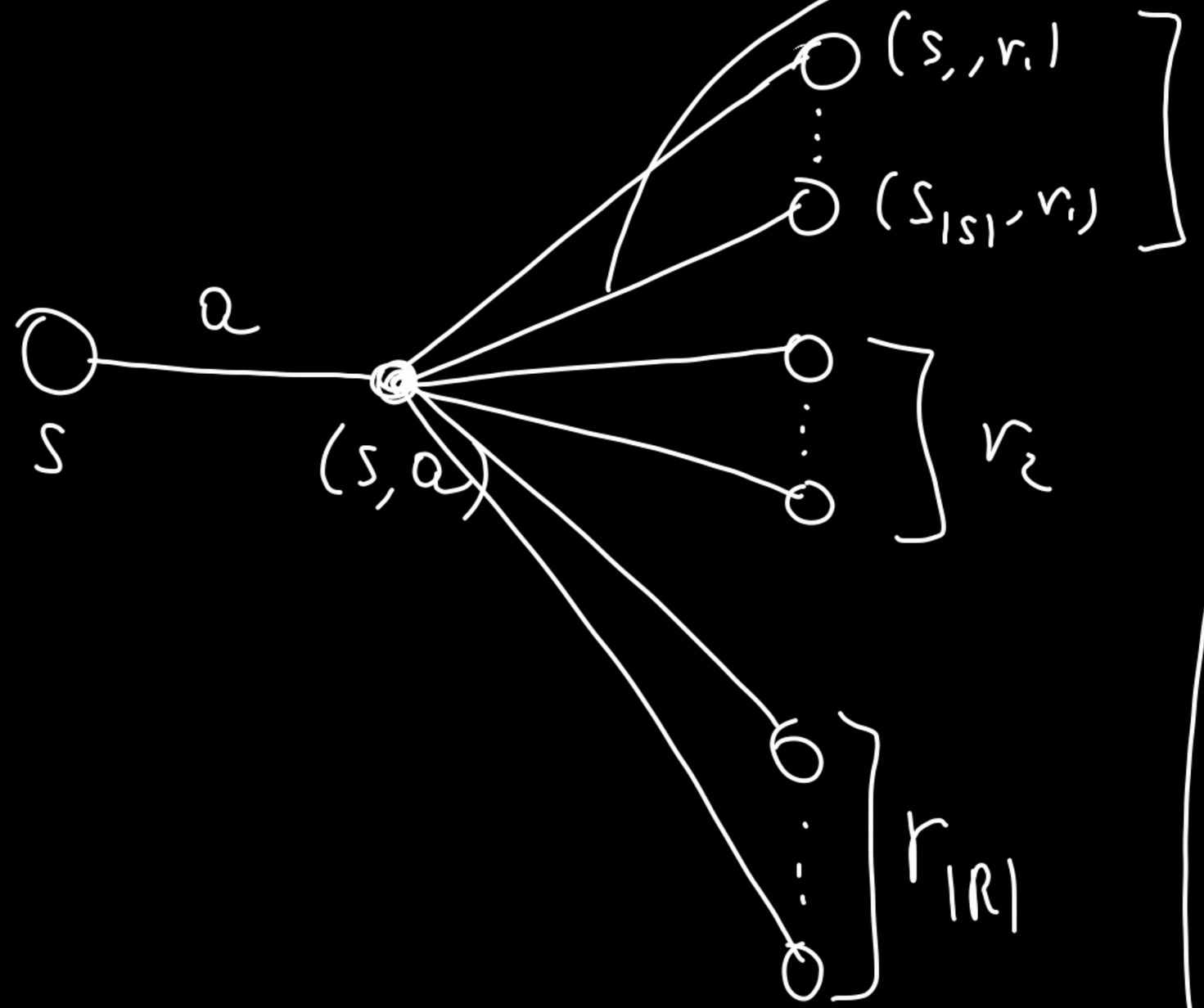
$p(s', r \mid s, a)$, FROM THIS, WE CAN COMPUTE SEVERAL OTHER USEFUL INFO:

- $p(s' \mid s, a) = \sum_{r \in R} p(s', r \mid s, a) \quad \forall s' \in S$

- $r(s, a) = E\{R_t \mid S_{t-1} = s, A_{t-1} = a\}$ (EXPECTED VALUE OF REWARD WHEN STARTING FROM s AND TAKING ACTION a)

COMPUTATION OF $r(s,a)$

$p(s', r | s, a)$



$p(r_1 | s, a)$

$E\{Y\}$

$Y \in \{y_1, y_2, y_3\}$

$P(y_1), P(y_2), P(y_3)$

$$E\{Y\} = \sum_i y_i \cdot P(y_i)$$

$$r(s,a) = E\{R_t \mid S_{t-1}=s, A_{t-1}=a\} =$$

TABLE

$$= \sum_{r\in R} r \cdot \underbrace{p(r|s,a)}_{\text{II}} = \sum_{r\in R} r \cdot \sum_{s'\in S} \underbrace{p(s',r|s,a)}$$

$$\sum_{s'\in S} p(s',r|s,a).$$

MARKOV PROPERTY

$$S_{t-1} \xrightarrow{P} S_t \xrightarrow{P} S_{t+1}$$

EXAMPLE: CLEANING ROBOT TO COLLECT TRASH.

$S = \{ \text{HIGH}, \text{LOW} \}$ (BATTERY LEVEL)

$A = \{ \text{SEARCH}, \text{WAIT}, \text{RECHARGE} \}$

$R = \{ r_{SEARCH}, r_{WAIT}, 0, -2 \}$

$(r_{SEARCH} > r_{WAIT})$

0 ↔ RECHARGE

-2 ↔ DISCHARGED WHILE SEARCHING

PROBABILITIES ($p(s'|s,a)$

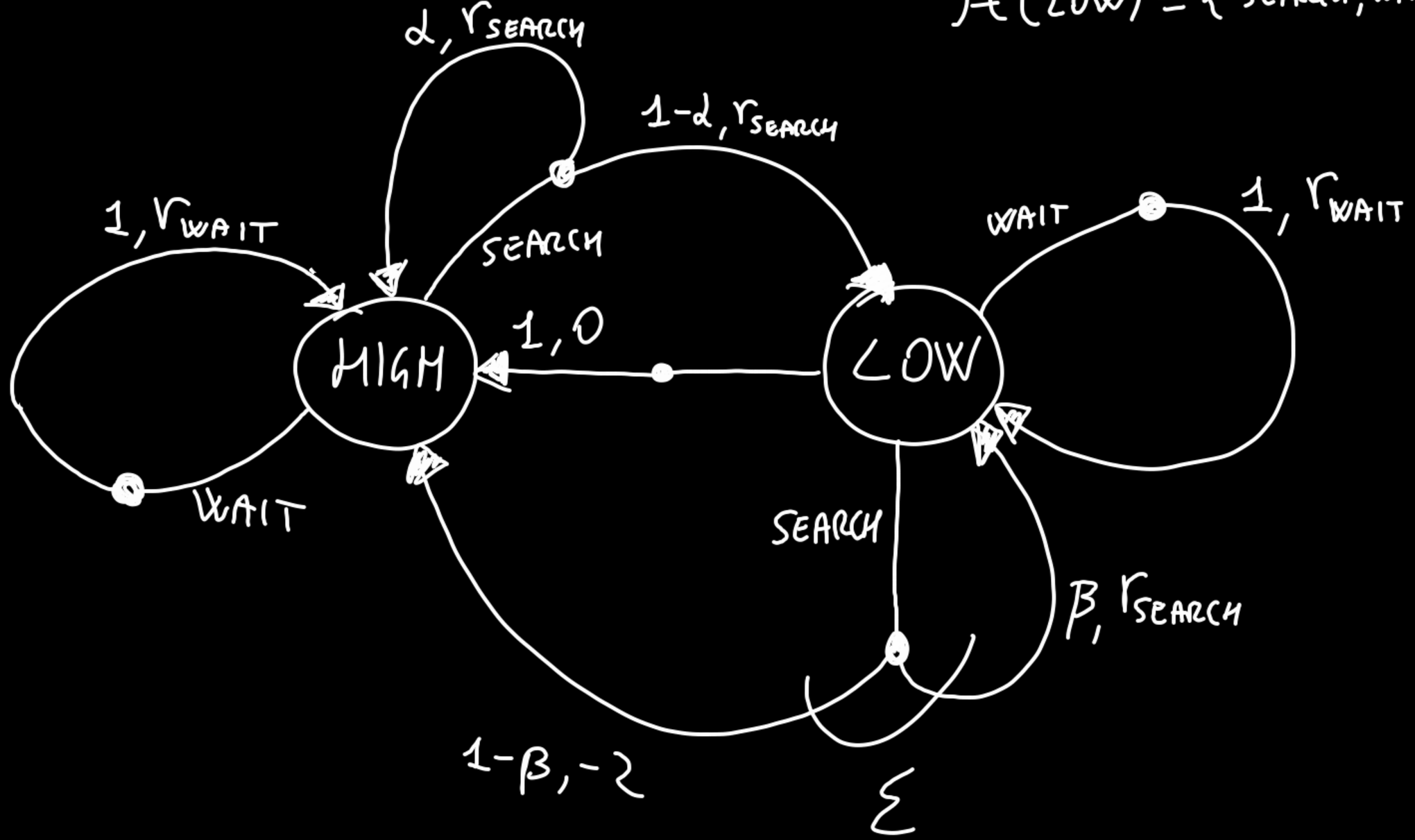
α: PROB. BATTERY AT $t+1$ IS HIGH, WHEN BATTERY AT t IS HIGH, AND THE AGENT DECIDES TO SEARCH

β: " " " LOW, " " " ,
" LOW, " " " .

TRANSITION GRAPH

$A(HIGH) = \{SEARCH, WAIT\}$

$A(LOW) = \{SEARCH, WAIT, RECHARGE\}$



DEFINITION OF V, Q AND ASSOCIATED PROPERTIES

S_t, R_t, A_t $\quad r_0, r_1, r_2 \ldots, r_T$ (EPISODIC)

$r_0, r_1, \ldots$ (CONTINUING)

RETURN AT TIME t, G_t:

$$G_t = R_{t+1} + R_{t+2} + \cdots = \sum_{k=0}^{\infty} R_{t+k+1}$$

(G_t DEPENDS ON π, $p(s', r|s, a)$ ←)

G_t (RETURN AT t IS THE SUM OF THE FUTURE REWARDS)

(DISCOUNTED) RETURN

$\gamma \in [0,1]$ DISCOUNT FACTOR $(\varepsilon ; \alpha ; \gamma)$

- ε: EXPLORATION RATE
- α: LEARNING RATE
- γ: DISCOUNT FACTOR

$$G_t = \sum_{k=0}^{\infty} \gamma^k R_{t+k+1} = 1 R_{t+1} + \gamma R_{t+2} + \gamma^2 R_{t+3} + \dots$$

G_t IS A CONVERGING SERIES (IF REWARDS ARE BOUNDED).

G_t TAKES FINITE VALUES.

- γ SMALL $\rightarrow$ SHORT-TERM PERFORMANCE (GREEDY AGENT)
- γ LARGE $\rightarrow$ LONG-TERM PERFORMANCE

SIMPLE ITERATIVE RECATION FOR G_t

$$G_t = R_{t+1} + \underbrace{\gamma R_{t+2} + \gamma^2 R_{t+3} + \gamma^3 R_{t+4} + \cdots}$$

$$G_t = R_{t+1} + \gamma \underbrace{\left(R_{t+2} + \gamma R_{t+3} + \gamma^2 R_{t+4} + \cdots \right)}$$

$$= \text{(DEFINITION OF } G_t \text{)}$$

$$\boxed{G_t = R_{t+1} + \gamma G_{t+1}}$$

$\pi(a|s)$ PROBABILITY OF SELECTING ACTION a

WHEN IN STATE s.

$$\pi(a|s) \triangleq \Pr\{A_t = a \mid S_{t-1} = s\}$$

POLICY

VALUE FUNCTIONS

- $V_\pi(s)$ – STATE VALUE FUNCTION
- $Q_\pi(s, a)$ – STATE-ACTION VALUE FUNCTION

DEF. OF $V_\pi(s)$ (DISCOUNTED) $V_\pi : \mathcal{S} \to \mathbb{R}$. $V_\pi(s)$ IS THE EXPECTED FUTURE RETURN OBTAINED WHEN STARTING FROM STATE s, AND FOLLOWING POLICY π THEREAFTER.

$$V_\pi(s) \triangleq E_\pi\{ G_t \mid S_{t-1} = s \} = E_\pi\{ \overbrace{R_{t+1} + \gamma G_{t+1}}^{G_t} \mid S_{t-1} = s \}$$

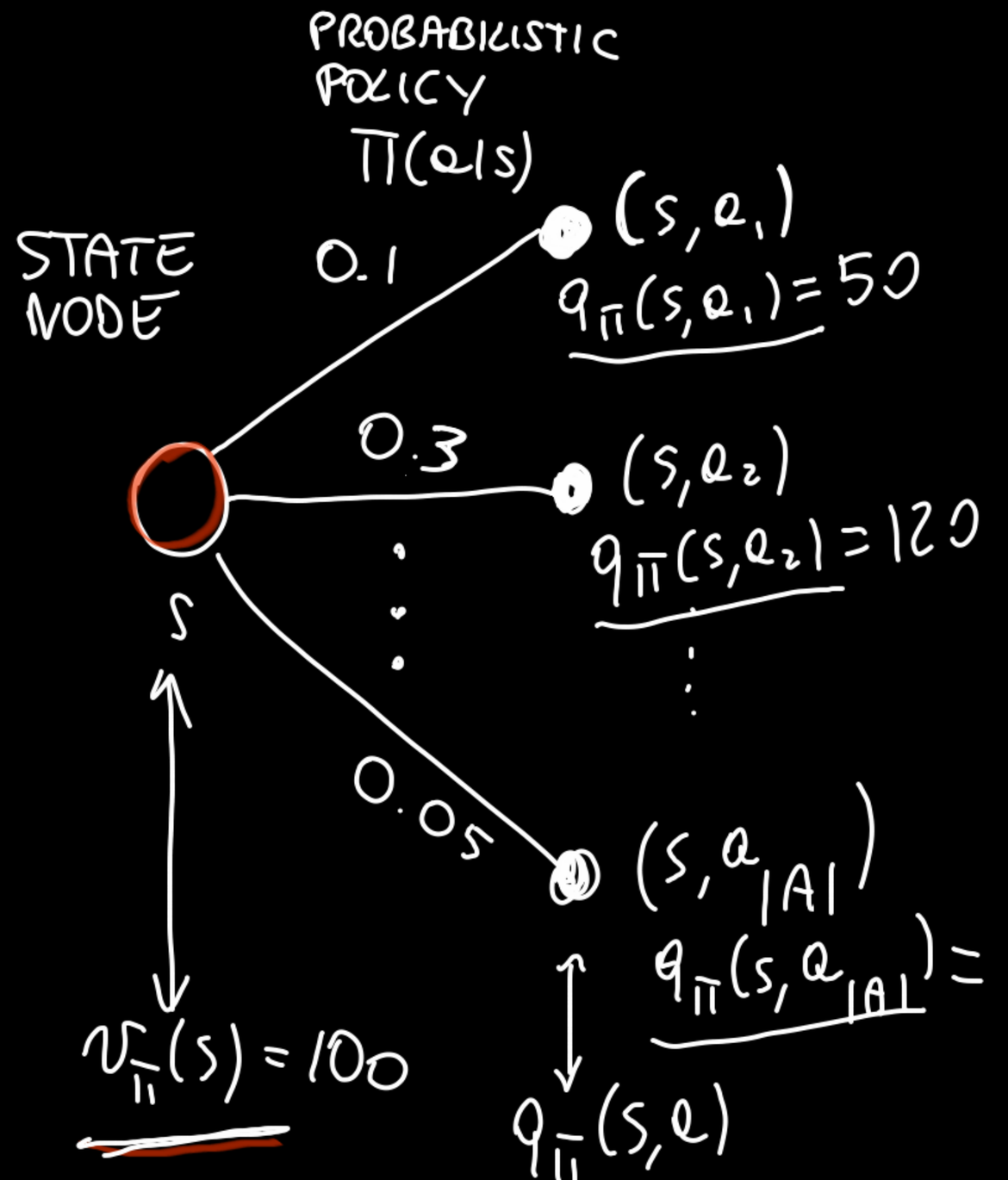
DEF. $q_\pi(s,a)$ $\quad q_\pi : \mathcal{S} \times \mathcal{A} \longrightarrow \mathbb{R}$ EXPECTED FUTURE DISC. RETURN WHEN ENVIRONMENT IS IN s, AGENT SELECTS a, AND FOLLOWS π THEREAFTER.

$$q_\pi(s,a) \triangleq E\{ G_t \mid S_{t-1} = s, A_{t-1} = a \}$$

"HOW GOOD IS A GIVEN ACTION FROM A GIVEN STATE, FOLLOWING A GIVEN POLICY."

(WHAT IS THE EXPECTED RETURN)

RELATION BETWEEN $v_\pi(s)$ AND $q_\pi(s,a)$

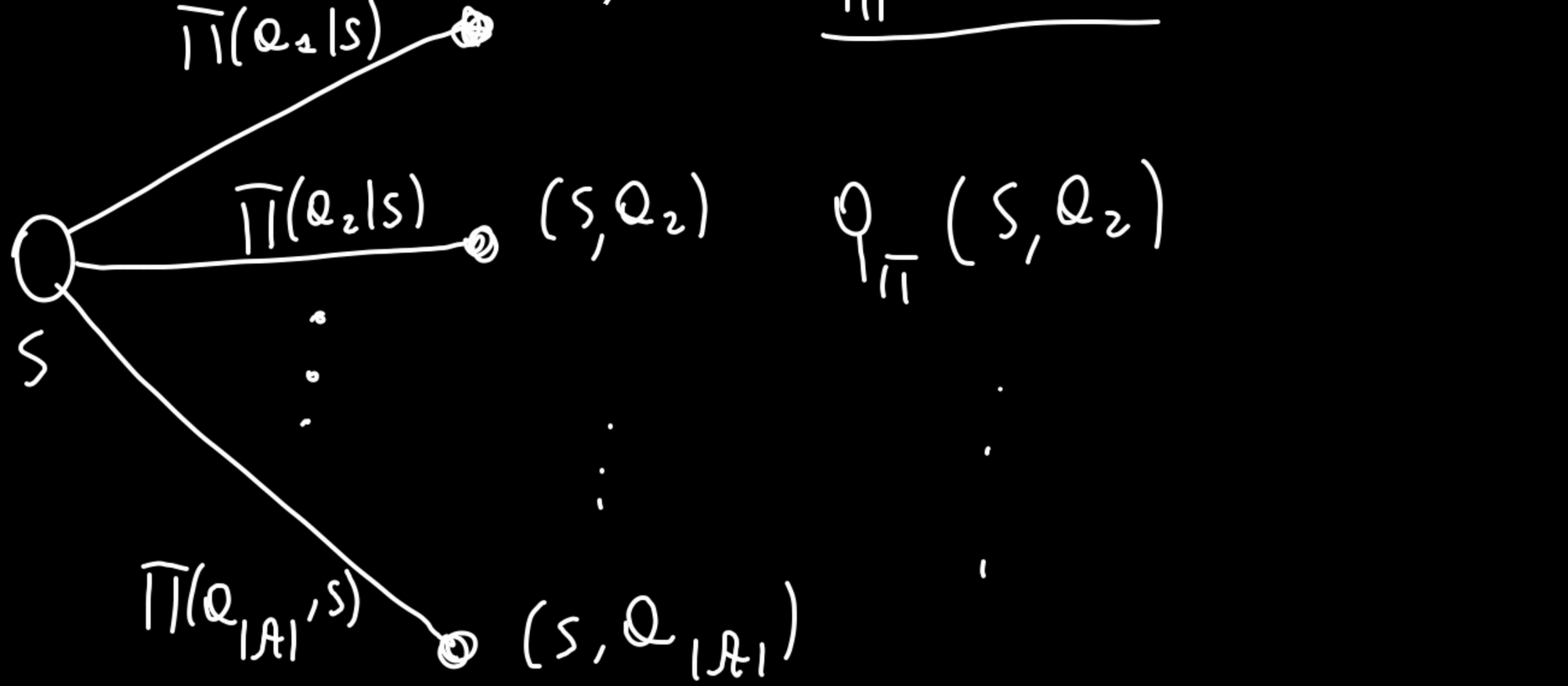


$v_\pi(s) = ?\left(q_\pi(s,a)\right)$

$v_\pi(s)$ IS EXP. VALUE OF FUTURE RETURN STARTING FROM s.

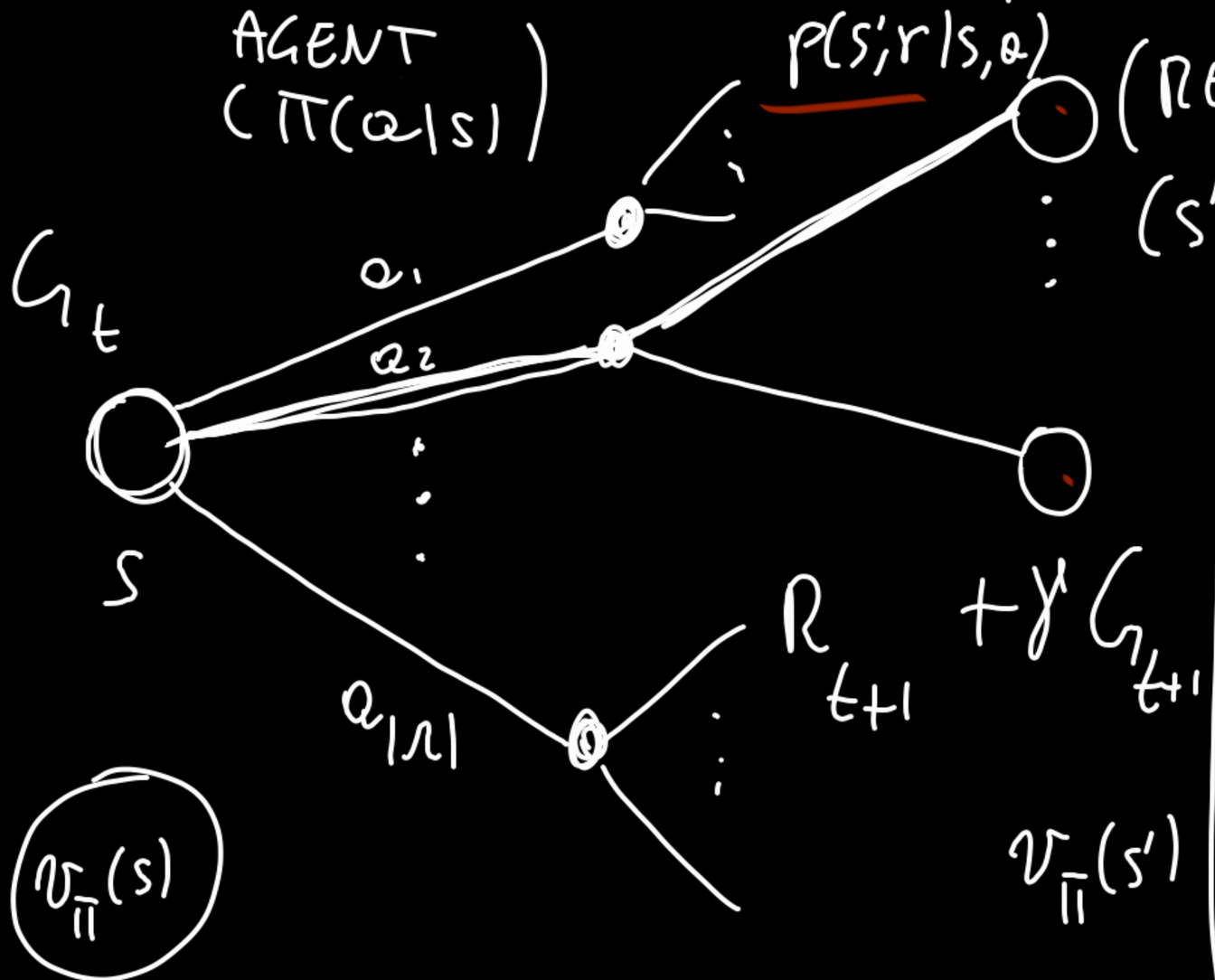
$$v_\pi(s) = \sum_{a \in A(s)} \pi(a|s) \cdot q_\pi(s,a)$$

(NO ENVIRONMENT P)



v_π AND q_π SATISFY A CONDITION CALLED BELLMANN EQ.

BELLMANN EQ. FOR $v_\pi(s) \triangleq E_\pi\{G_t \mid S_{t-1} = s\}$



$$v_\pi(s) \triangleq E_\pi\{R_{t+1} + \gamma G_{t+1} \mid S_{t-1} =$$

